

Sales and Chinook Reports

Project Objective

The objective of this project was to transition from flat-file data processing to managing advanced relational database architectures and multi-table querying frameworks. The project focused on deploying real-world relational schemas, executing advanced multi-table joins, writing aggregate subqueries, and implementing conditional metrics to extract precise business intelligence, evaluate global market structures, and solve strategic business-driven questions using SQL.

Key Findings

Enterprise Sales Platform Insights

- **Classic Cars Pricing Power:** Aggregation metrics isolate Classic Cars as the primary commercial driver, contributing a dominant \$3,919,615.66 to gross sales with 33,992 total units sold across 199 orders. This performance is anchored by strong inventory pricing power, commanding the highest average unit price across the enterprise at \$115.38.
- **Niche Capital Allocation:** Low-performing product categories like Trains contribute minimal lifetime revenue (\$226,243.47 across 45 orders) compared to the core automotive divisions. This significant gap highlights an immediate opportunity to streamline slow-moving stock and reallocate manufacturing capital into higher-margin vehicle lines.
- **Territorial Revenue Concentration:** International business stability relies heavily on an elite tier of global regions that cross the \$500,000 lifetime milestone. As proven by the analytical filtering query, the United States serves as the primary operational anchor (\$3,627,982.83), while Spain (\$1,215,686.92) and France (\$1,110,916.52) form a high-yield European core, followed by Australia (\$630,623.10).
- **Macro Annual Revenue Trajectory:** Grouped aggregate computations executed provide a clear view of enterprise macro scaling across active operational periods. Initial fiscal structures established a baseline baseline in 2003 with \$3,516,979.54 in total revenue, which experienced a massive 34.3% surge to peak at \$4,724,162.60 in 2004. Transactions recorded for 2005 stand at \$1,791,486.71, representing a partial-year register capturing active record before final dataset compilation. The dataset's operational lifecycle spans from a start date of 2003-01-06 to a terminal log of 2005-05-31. This boundary verification conclusively proves that the 2005 metric represents a partial 5-month operational window rather than a macroeconomic commercial contraction.

Chinook Music Store Insights

- **Legacy Rock Dominance:** Transaction volume is heavily concentrated in heavy metal and hard rock genres. A small group of classic artists, led by Iron Maiden (\$138.60 revenue; 140 tracks sold) and U2 (\$105.93 revenue; 107 tracks sold), drives the vast majority of catalog sales.
- **Uniform Purchasing Behavior:** Despite major differences in total transaction volumes across global regions, the multi-join query tracking order reveals that the average number of tracks purchased per invoice remains completely static at exactly 5.43 items per order across all primary countries (USA, Canada, Brazil, France, Germany, United Kingdom, and Czech Republic). This horizontal baseline strongly suggests that consumer cart depth is heavily shaped by digital UI constraints or uniform basket packaging rather than localized taste variations.
- **Premium Asset Conversions:** High-definition video and long-form television tracks successfully clear the store's baseline price average. The advanced subquery isolates active assets priced at a premium \$1.99 (such as Occupation / Precipice, Exodus, Pt. 1 & 2, and Collaborators) that have high transaction validation, showing that users are willing to pay double the base price for long-form content.

Conclusion

This project demonstrates the core value of relational data engineering and advanced querying in transforming raw database tables into targeted corporate insights. By moving beyond flat-file extraction into complex multi-table structures, aggregate filtering, and relational database auditing, the datasets were successfully shaped into highly reliable assets.